

Architecture Design Document

High-Level Architecture Diagram

External APIs (Yahoo Finance, Alpha Vantage) → Apache Airflow → ETL Services → PostgreSQL & Neo4j → AI Services → Streamlit UI

End-to-End Workflow Diagram

Airflow Scheduler → Stock ETL → PostgreSQL → News ETL → Sentiment Analysis → Research Scoring → Risk Scoring → Streamlit Dashboard

Data Flow Architecture

Yahoo Finance → Stock ETL → stock_prices; Alpha Vantage → News ETL → news_metadata; Sentiment Service → sentiment_scores; Research Service → research_scores; Risk Service → risk_scores

Agent Architecture

User Query → Research Agent → Stock Agent / News Agent / Risk Agent → Databases → Final Response

Deployment Architecture

Docker Environment containing Airflow, PostgreSQL, Neo4j, and Streamlit. Airflow executes ETL jobs and stores results in databases for dashboard consumption.